Topic 11 – Static electricity

Students should:		Maths skills
11.1P	Explain how an insulator can be charged by friction, through the transfer of electrons	
11.2P	Explain how the material gaining electrons becomes negatively charged and the material losing electrons is left with an equal positive charge	
11.3P	Recall that like charges repel and unlike charges attract	
11.4P	Explain common electrostatic phenomena in terms of movement of electrons, including a shocks from everyday objects b lightning c attraction by induction such as a charged balloon attracted to a wall and a charged comb picking up small pieces of paper	
11.5P	Explain how earthing removes excess charge by movement of electrons	
11.6P	Explain some of the uses of electrostatic charges in everyday situations, including insecticide sprayers	
11.7P	Describe some of the dangers of sparking in everyday situations, including fuelling cars, and explain the use of earthing to prevent dangerous build-up of charge	
11.8P	Define an electric field as the region where an electric charge experiences a force	
11.9P	Describe the shape and direction of the electric field around a point charge and between parallel plates and relate the strength of the field to the concentration of lines	5b
11.10P	Explain how the concept of an electric field helps to explain the phenomena of static electricity	

Suggested practicals

- Investigate the forces of attraction and repulsion between charged objects.